

TruHealth

Report

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Welcome to your personalized report on Epigenetic Biomarkers

What are Epigenetic Biomarkers?

Epigenetic biomarkers are specific biological markers derived from analyzing DNA methylation patterns in your genome. Unlike traditional blood-based metrics of your health such as serum levels, which provide a single snapshot in time, epigenetic biomarkers uniquely offer insights into ongoing biological functions by predicting levels of nutrients, metabolites, and other essential biochemical entities. These epigenetic measurements are based on patterns in your DNA methylation landscape, influenced by both genetic and environmental factors. Epigenetic biomarkers provide a more stable and comprehensive view of your health over time, reflecting long-term exposures and trends rather than short-term fluctuations.

Epigenetic Biomarkers vs Serum Levels

Serum levels represent a direct measurement of substances in your blood at a specific point in time. These levels can vary based on recent meals, exercise, stress, and other transient factors. In contrast, epigenetic biomarkers are derived from DNA methylation patterns, providing a broader and more consistent indication of health, which isn't as influenced by short-term changes. This makes epigenetic biomarkers especially useful for understanding long-term health trends and risks.

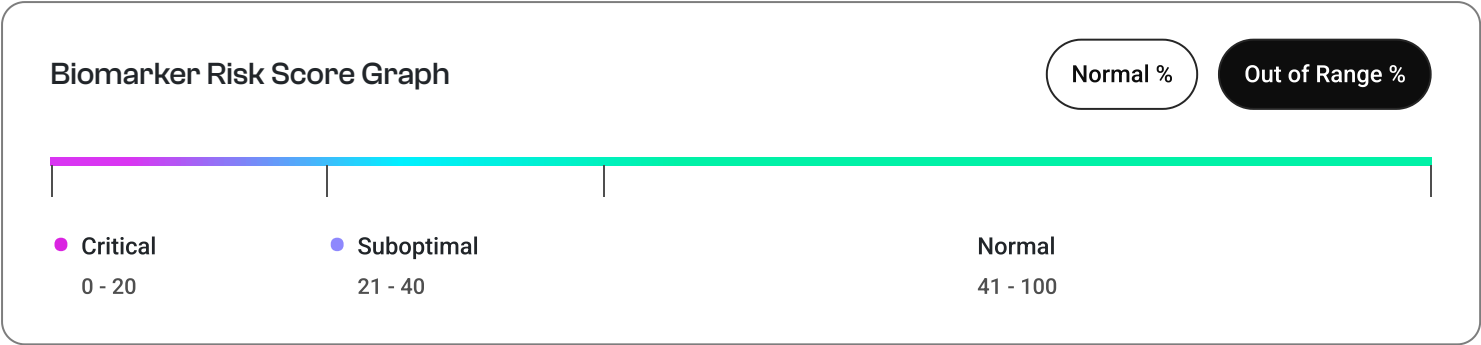
Epigenetic biomarkers provide a more stable and comprehensive view of your health over time, reflecting long-term exposures and trends rather than short-term fluctuations.

The Reference Cohort: The Data Behind TruHealth's Analytics

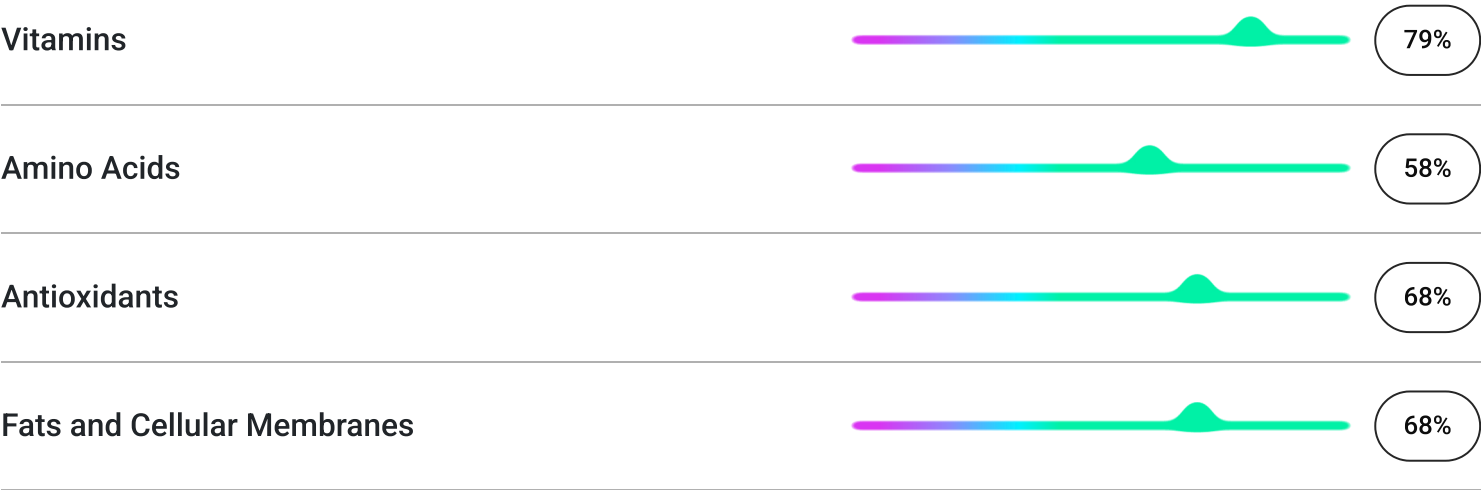
Our algorithms and biomarkers are developed and validated using data from a subset of the Massachusetts General Brigham (MGB) Biobank. The MGB Biobank is a comprehensive repository containing over 130,000 high-quality samples from more than 100,000 consented patients. These samples are linked to detailed EMR data, which includes the patients' lifetime medical histories. The biobank also incorporates survey data on lifestyle, environment, and family history. To ensure meaningful percentile comparisons in your report, we used an equal combination of data from the MGB Biobank and TruDiagnostic patients, providing a robust, diverse, and extensively studied reference population.

Biomarkers Overview

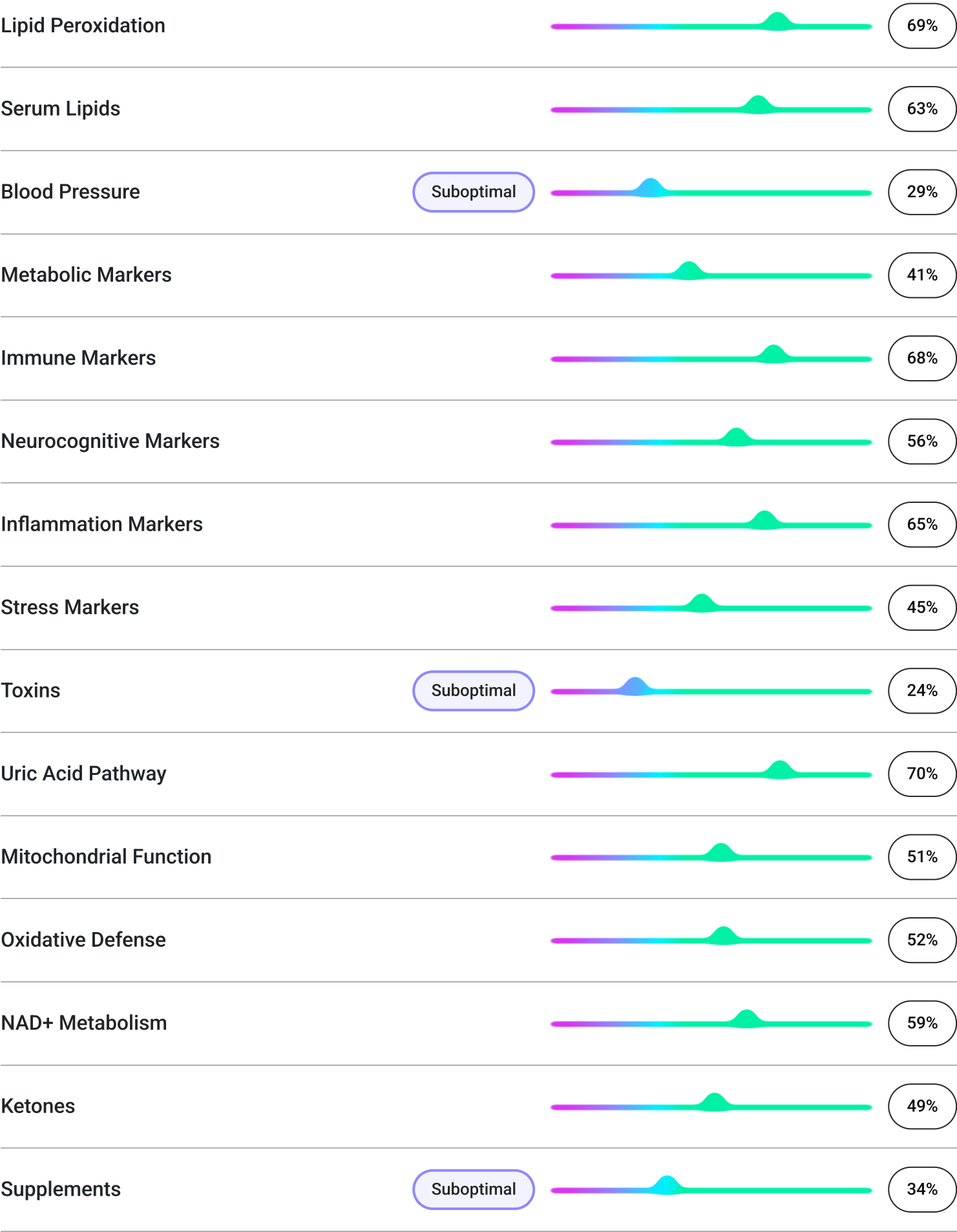
This section of the TruHealth report provides a comprehensive overview of an individual's general health and nutrition markers, each derived from one or more epigenetic biomarkers that reflect underlying cellular health and function. These markers are presented in a normalized percentile range, allowing for a straightforward comparison with a reference population. A score of 0 indicates the worst outcome, representing potential areas of concern or risk, while a score of 100 signifies the best possible outcome, highlighting areas of optimal health. The percentile ranking provides an easy-to-interpret indication of how an individual's health status compares to the broader population, offering valuable insights into their relative standing in terms of metabolic health, immune function, inflammation, mitochondrial activity, and more. This comparison against a carefully studied cohort gives context to these markers, enabling users to understand areas where they may need intervention or where they are already performing well.



NUTRITION



GENERAL HEALTH MARKERS



TOP PERSONALIZED RECOMMENDATIONS

SERUM LIPIDS

LDL-C

Lower LDL-C by eating a heart-healthy diet (rich in fiber, omega-3s, and healthy fats), avoiding trans fats and added sugars, exercising regularly, and maintaining a healthy weight.

🔗 Red Yeast Rice

🔗 Omega 3's

NEUROCOGNITIVE MARKERS

Dopamine Metabolites

Manage stress, get quality sleep, engage in moderate exercise, and reduce stimulants like caffeine.

🔗 L-theanine

🔗 Magnesium

TOXINS

PFAS (Forever Chemicals)

Strict avoidance of contaminated water and food.

🔗 Activated charcoal

🔗 N-acetylcysteine (NAC)

OXIDATIVE DEFENSE

Dimethyllysine

Mediterranean eating, fewer ultra-processed foods and sugars, regular exercise, sleep, tobacco avoidance

🔗 Omega-3s

🔗 Benfotiamine or carnosine with clinician

BLOOD PRESSURE

Dimethylarginine, SDMA and ADMA

Aerobic plus resistance training, Mediterranean diet, manage BP, lipids, and glucose, weight loss if needed

🔗 EPA or DHA

🔗 Cocoa flavanols

METABOLIC MARKERS

Satiety Hormone

Reduce body fat through a balanced diet rich in whole foods, moderate caloric restriction, and strength training and aerobic activities. Improving sleep quality.

🔗 Berberine

🔗 Omega-3s

EPIGENETIC BIOMARKERS RISK SCORES

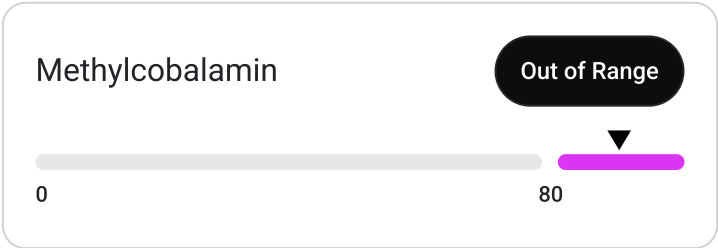
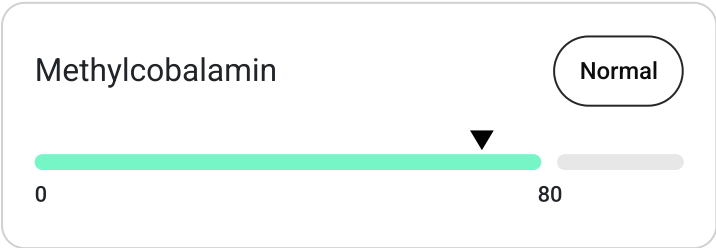
The **Epigenetic Biomarkers Risk Scores** section of the report provides a detailed breakdown of specific **epigenetic biomarkers** derived from your epigenome, each linked to relevant **risk scores** and associated **biomarkers**. Every epigenetic biomarker is accompanied by a concise explanation of its function in the body and its relationship to disease or health outcomes. The **optimal and suboptimal ranges** for these biomarkers vary, depending on whether the biomarker is most beneficial at **high, low, or midrange levels**. For biomarkers identified as suboptimal, personalized **lifestyle** and **supplement recommendations** are provided for improvement. Additionally, prior epigenetic biomarker values are displayed to **track changes over time**, offering insight into progress and areas that may need continued attention.

There are Three Different Ranges for Each Epigenetic Biomarkers (EB)

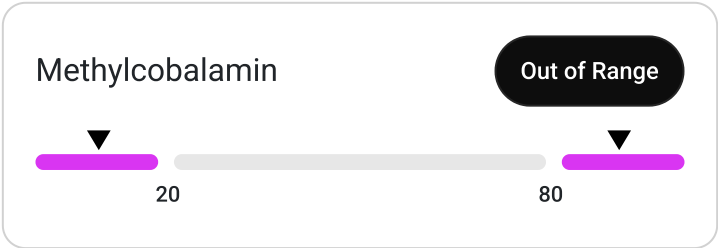
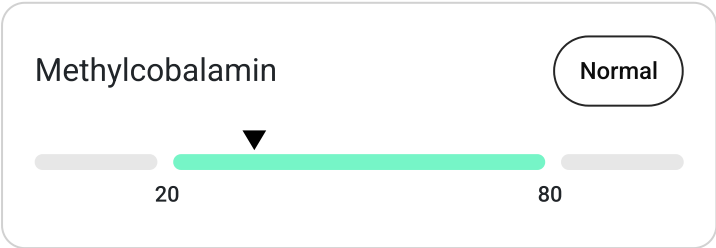
In the below section, each epigenetic biomarker will be presented as your percentile when compared against the reference population. Each epigenetic biomarker is identified by our science team as ideal when LOW, MIDDLE, or HIGH. When the ideal biomarker is thought to be LOW, a percentile of 0% is best. When the MIDDLE is ideal, 50% is best. When HIGH is ideal, 100% is best.

Normal Out of Range ▲ Critically Out of Range

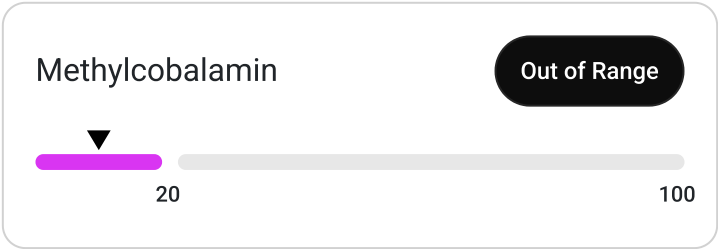
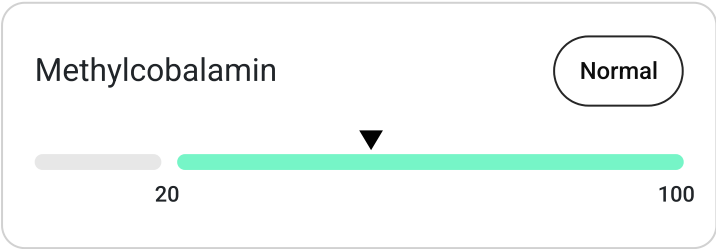
Example: Normal Range is **LOW**



Example: Normal Range is **MIDDLE**



Example: Normal Range is **HIGH**



VITAMINS

Vitamin A_m

Retinol (vitamin A)_m



Description

Retinol (Vitamin A) is essential for vision, immune function, and cellular communication.

Vitamin B2 (Riboflavin)_m

Riboflavin (vitamin B2)_m



Description

A B-vitamin essential for energy production and cellular function

Vitamin B3 (Nicotinamide)_m

Nicotinamide_m



Description

Nicotinamide is a form of vitamin B3, essential for NAD+ production and cellular energy. Low levels are associated with lower NAD+ production, and high levels are associated with inhibition of Sirtuins and NAD+ recycling pathways.

Vitamin B5 (Pantothenic Acid)_m

Pantothenic Acid (vitamin B5)_m



Description

Pantothenic Acid (Vitamin B5) is essential for CoA synthesis and energy metabolism.

Vitamin B6 (P5P)_m

Pyridoxine, Pyridoxine-5-Phosphate (vitamin B6)_m



Description

Pyridoxine (Vitamin B6) is a coenzyme in amino acid metabolism and neurotransmitter synthesis.

Vitamin B8 (Inositol)_m

Inositol (myoinositol)_m



Description

Inositol is a carbohydrate involved in cell membrane formation and insulin signal transduction.

Vitamin C_m

Ascorbic acid 2-sulfate_m

5

100

86%

Description

Sulfated form of Vitamin C.

Vitamin C_m

Ascorbic acid 3-sulfate_m

5

100

57%

Description

Another sulfated form of Vitamin C.

Vitamin D_m

Cholecalciferol (vitamin D)_m

20

98

81%

Description

A form of Vitamin D; important for bone health and immune function.

Vitamin E_m

Alpha-tocopherol_m

10

98

57%

Description

A form of Vitamin E with strong antioxidant properties. High levels can be associated with reduced gamma-tocopherol function, and low levels can be associated with poor antioxidant function.

Choline_m

Total cholines_m

10

100

79%

Description

Essential nutrient involved in brain health and fat metabolism.

Betaine_m

5

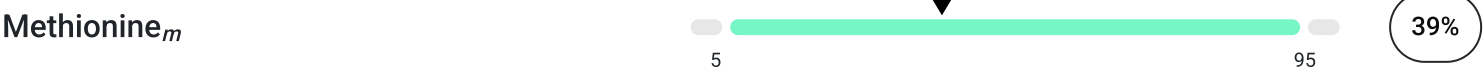
100

81%

Description

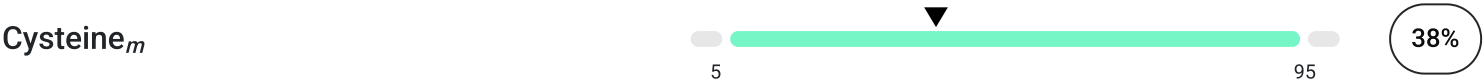
A methyl donor involved in liver function and cell hydration.

AMINO ACIDS



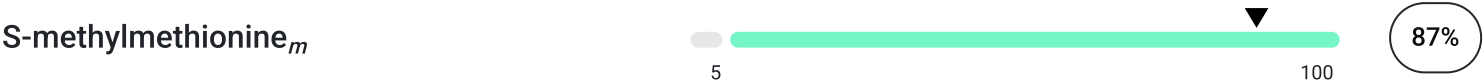
Description

An essential amino acid involved in protein synthesis and detoxification processes.



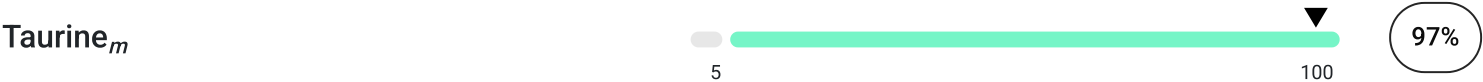
Description

Amino acid involved in protein synthesis and antioxidant functions



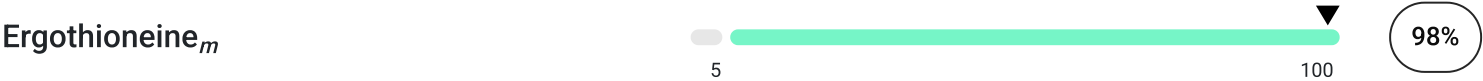
Description

A methylated form of methionine involved in methylation reactions



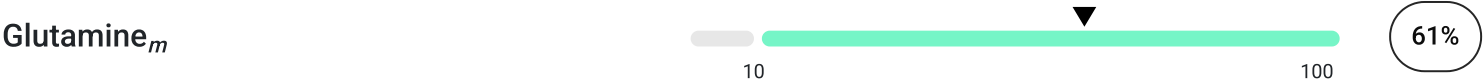
Description

A sulfur-containing amino acid with antioxidant properties.



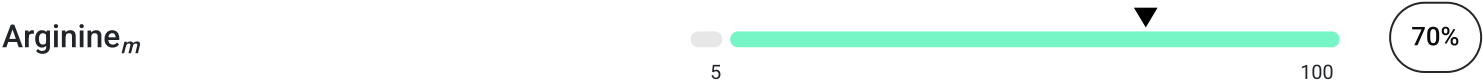
Description

Ergothioneine is a powerful antioxidant involved in cellular protection and oxidative stress regulation. Low levels could indicate reduced cellular defense against oxidative damage.



Description

An amino acid essential for gut health.



Description

An amino acid involved in protein synthesis and nitric oxide production.

L-Aspartic Acid_m



Description

L-Aspartic Acid is a non-essential amino acid important in the urea cycle and energy production. Low levels may impair protein synthesis and neurotransmitter function, while high levels could indicate metabolic stress.

Valine_m

N-carbamoylvaline_m



Description

N-carbamoylvaline is a derivative of valine, playing a role in nitrogen metabolism. Abnormal levels might indicate stress on nitrogen metabolism from diet or inflammation.

Asparagine_m



Description

An amino acid involved in metabolic processes.

Threonine_m

N-acetylthreonine_m



Description

N-acetylthreonine is an acetylated derivative of threonine, essential for protein production. Low levels could impair immune function and recovery, while high levels might suggest dietary imbalances.

Valine_m

N-acetylvaline_m



Description

N-acetylvaline is involved in valine metabolism, an essential amino acid. Low levels could affect muscle recovery, while high levels could reflect metabolic stress.

Glycine_m



Description

An amino acid involved in detox and sleep.

Carnosine_m

N-acetylcarnosine_m

High

090

91%

Description

A derivative of carnosine, which acts as an antioxidant in the body.

Recommendations

Limit red and processed meats, prioritize antioxidant-rich foods like berries and leafy greens, regular exercise with adequate rest to reduce muscle breakdown.

- Omega 3s
- NAC
- CoQ10

Cystathionine_m

590

83%

Description

Intermediate in methionine metabolism and cysteine biosynthesis

Histidine_m

N-acetylhistidine_m

High

590

90%

Description

A derivative of histidine, important for metal ion binding and antioxidant functions. Low levels could affect muscle recovery, while high levels may indicate inflammation or stress.

Recommendations

Avoid excess histidine supplementation and monitor for imbalances.

- Limit histidine supplements

Citrulline_m

5100

31%

Description

Non-essential amino acid involved in nitric oxide production and vascular health.

ANTIOXIDANTS

Carotenoids_m

Lutein_m

5100

47%

Description

A carotenoid with antioxidant properties, found in green leafy vegetables.

Carotenoids_m

Carotene diol_m

10

98

81%

Description

Carotenoid found in plants; antioxidant properties.

Acetyl-L-Carnitine_m

5

100

11%

Description

Acetyl-L-Carnitine is a compound involved in fatty acid metabolism and mitochondrial energy production. Low levels could suggest impaired mitochondrial function or reduced fatty acid utilization.

Vitamin A_m

Retinol (vitamin A)_m

10

98

98%

Description

Retinol (Vitamin A) is essential for vision, immune function, and cellular communication.

Vitamin E_m

Alpha-tocopherol_m

10

98

57%

Description

A form of Vitamin E with strong antioxidant properties.

Ergothioneine_m

5

100

98%

Description

Ergothioneine is a powerful antioxidant involved in cellular protection and oxidative stress regulation. Low levels could indicate reduced cellular defense against oxidative damage.

Vitamin B2 (Riboflavin)_m

Riboflavin (vitamin B2)_m

5

100

74%

Description

A B-vitamin essential for energy production and cellular function

Vitamin C_m

Ascorbic acid 2-sulfate_m

5

100

86%

Description

A sulfated form of Vitamin C.

Vitamin C_m

Ascorbic acid 3-sulfate_m

5

100

57%

Description

Another sulfated form of Vitamin C.

FATS AND CELLULAR MEMBRANES

Omega 3_m

Omega-3_m

20

100

86%

Description

Omega-3 is a polyunsaturated fatty acid with essential roles in reducing inflammation and supporting heart, brain, and joint health. High levels may indicate adequate dietary intake, while low levels could suggest poor dietary consumption or increased risk of inflammatory conditions.

DHA_m

Docosahexaenoate (DHA; 22:6n3)_m

10

100

62%

Description

An omega-3 fatty acid that plays a key role in brain health and inflammation.

DPA_m

Docosapentaenoic acid_m

10

100

13%

Description

A lesser-known omega-3 fatty acid involved in inflammatory processes.

EPA_m

Eicosapentaenoate (EPA; 20:5n3)_m

10

100

48%

Description

An omega-3 fatty acid with anti-inflammatory effects, found in fish oils.

Omega 6_m

Omega-6_m

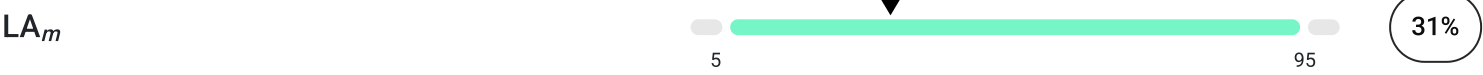
10

100

57%

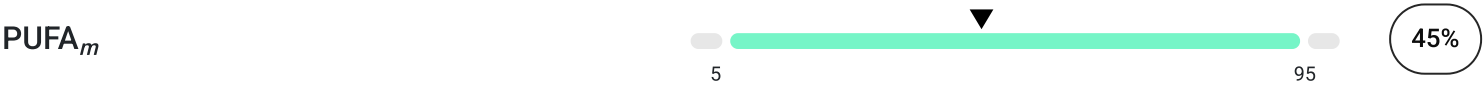
Description

Omega-6 is a polyunsaturated fatty acid that supports cell membrane integrity and helps regulate inflammatory processes. Elevated levels may reflect a diet rich in whole-food sources (e.g., nuts, seeds, plant oils) and have been linked with lower all-cause and cardiovascular mortality. Low levels could indicate inadequate intake of essential fats or dysregulated lipid metabolism.



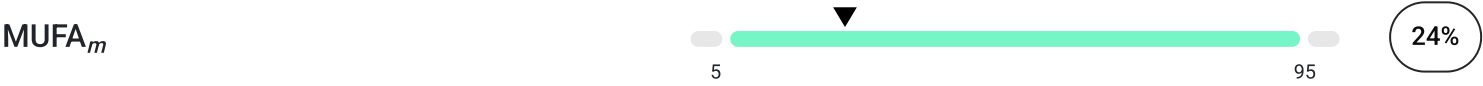
Description

Linoleic Acid (LA) is an essential omega-6 polyunsaturated fatty acid critical for cell membrane integrity and inflammatory processes. Excessive LA intake may promote lipid peroxidation and chronic inflammation, potentially negatively affecting metabolic health and longevity.



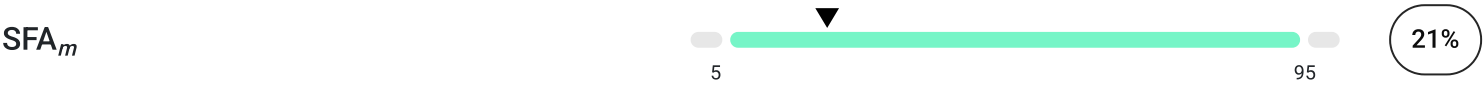
Description

Polyunsaturated Fatty Acids (PUFAs) are essential fats involved in cell structure and inflammation regulation. Excessive omega-6 PUFA intake may promote lipid peroxidation, leading to oxidative stress and inflammation, which could negatively impact longevity and metabolic health.



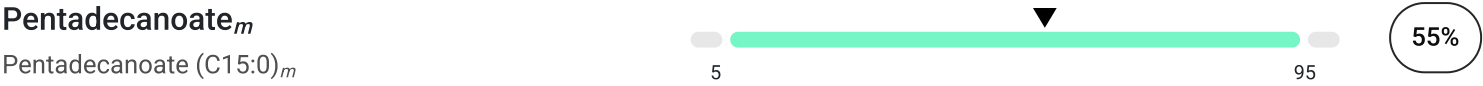
Description

Monounsaturated Fatty Acids (MUFAs) are healthy fats with a single double bond, involved in maintaining cell membrane fluidity and supporting cardiovascular health. MUFAs may reduce inflammation and oxidative stress, potentially benefiting metabolic health and longevity.



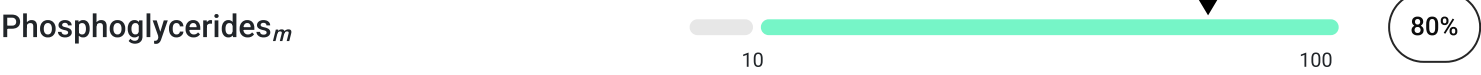
Description

Saturated Fatty Acids (SFAs) are fats with no double bonds, commonly found in animal products and some plant oils. High intake of SFAs can increase cholesterol levels and may promote inflammation, potentially impacting cardiovascular health and longevity when consumed in excess.



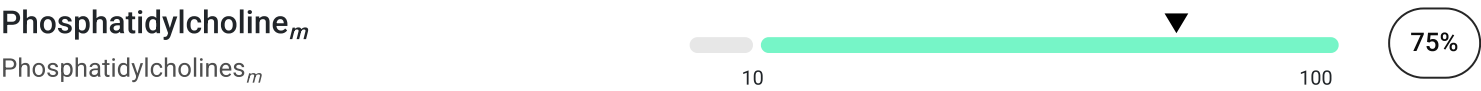
Description

Pentadecanoate (C15:0) is a saturated fatty acid with potential anti-inflammatory properties. High levels may indicate metabolic stress or excessive intake of specific dietary fats, while low levels could suggest impaired fat metabolism.



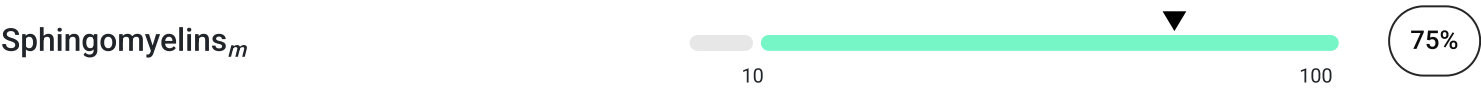
Description

Phosphoglycerides are a class of phospholipids that form a major component of cell membranes, supporting membrane fluidity, signaling, and energy metabolism. Imbalances in phosphoglycerides may disrupt cellular function, potentially affecting metabolic health and longevity.



Description

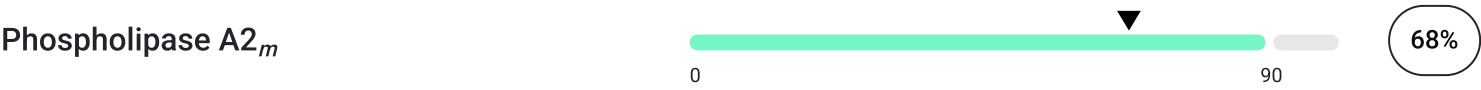
Phosphatidylcholines are a major class of phospholipids critical for cell membrane structure, lipid transport, and signaling. They support liver function and cognitive health, with imbalances potentially affecting metabolism and longevity.



Description

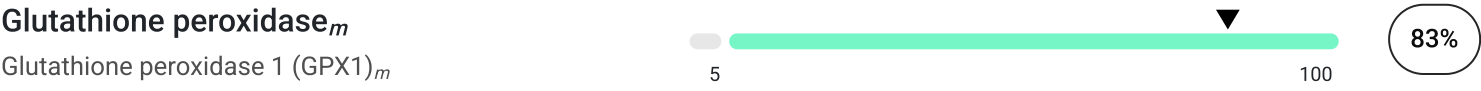
Sphingomyelins are a type of sphingolipid essential for cell membrane integrity and signaling, particularly in nerve and brain tissues. Dysregulation of sphingomyelins may contribute to metabolic dysfunction and neurodegenerative diseases, potentially impacting longevity.

LIPID PEROXIDATION



Description

Phospholipase A2 is an enzyme involved in lipid metabolism and inflammatory regulation. High levels may indicate heightened inflammatory activity or cellular damage, while low levels could suggest impaired lipid signaling or reduced inflammatory response.



Description

Glutathione peroxidase is an enzyme involved in antioxidant defense and cellular protection. Elevated levels may reflect increased oxidative stress or an adaptive response to cellular damage, while low levels could indicate impaired antioxidant capacity and increased vulnerability to oxidative damage.

Octadecadienedioate (C18:2-DC)_m



Description

C18:2-DC is a dicarboxylic acid, linked to lipid metabolism and energy regulation. High levels could be linked to metabolic disorders, while low levels may indicate impaired lipid metabolism or energy production.

SERUM LIPIDS

ApoB_m



Description

Apolipoprotein B (ApoB) is a structural protein of atherogenic lipoproteins, including LDL and VLDL. Elevated ApoB levels reflect the number of these lipoproteins, providing a stronger predictor of cardiovascular disease risk and atherosclerosis than LDL cholesterol alone, making it crucial for assessing metabolic health and longevity.

LDL-C_m



Description

LDL-C (Low-Density Lipoprotein Cholesterol) represents the cholesterol carried by LDL particles, often referred to as "bad cholesterol." Elevated LDL-C levels are strongly associated with increased risk of atherosclerosis and cardiovascular disease, making it a key marker for assessing cardiovascular health and longevity.

Recommendations

Lower LDL-C by eating a heart-healthy diet (rich in fiber, omega-3s, and healthy fats), avoiding trans fats and added sugars, exercising regularly, and maintaining a healthy weight.

- Red Yeast Rice
- Omega 3's
- Berberine
- Psyllium Husk

LDL particle size_m



Description

Smaller, dense LDL particles are more atherogenic and strongly associated with cardiovascular disease risk, while larger LDL particles are less likely to penetrate arterial walls, indicating a potentially lower risk profile and better metabolic health.



Description

VLDL-C (Very-Low-Density Lipoprotein Cholesterol) represents cholesterol carried by VLDL particles, which transport triglycerides from the liver to peripheral tissues. Elevated VLDL-C is associated with increased atherogenic risk, contributing to triglyceride-rich lipoprotein accumulation, cardiovascular disease, and metabolic dysfunction, negatively impacting longevity.



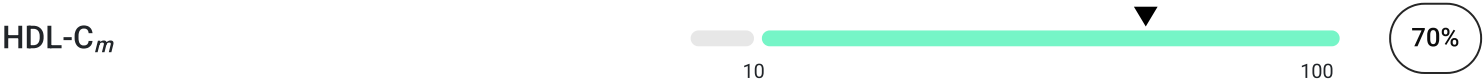
Description

Smaller VLDL particles are more atherogenic and linked to higher cardiovascular risk, while larger particles may indicate improved lipid metabolism and reduced metabolic dysfunction, influencing overall health and longevity.



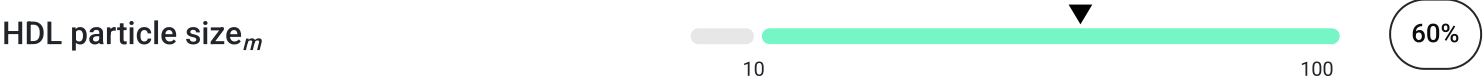
Description

Apolipoprotein A1 (ApoA1) is the primary structural protein of HDL particles, playing a key role in reverse cholesterol transport and antioxidant defense. Higher ApoA1 levels are associated with improved cardiovascular health, reduced atherosclerosis risk, and better longevity outcomes through enhanced lipid metabolism and anti-inflammatory effects.



Description

HDL-C (High-Density Lipoprotein Cholesterol) represents the cholesterol carried by HDL particles, often referred to as "good cholesterol." Higher HDL-C levels are associated with improved reverse cholesterol transport, reduced cardiovascular disease risk, and better longevity outcomes through anti-inflammatory and antioxidant effects.



Description

Larger HDL particles are generally more effective at reverse cholesterol transport and provide stronger antioxidant and anti-inflammatory benefits, contributing to reduced cardiovascular disease risk and improved longevity.



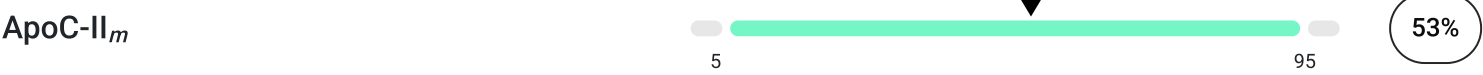
Description

Total triglycerides are the primary form of fat in the blood, stored for energy use in adipose tissue. Elevated levels are associated with insulin resistance, metabolic syndrome, and increased cardiovascular risk, making them a critical marker for assessing metabolic health and longevity.



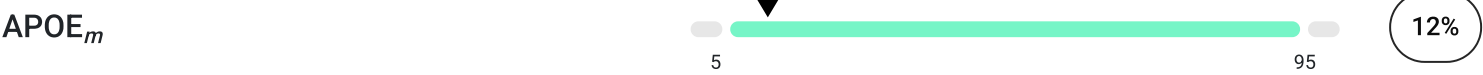
Description

Cholesterol carried in triglyceride-rich particles. Higher levels associated with causal ASCVD risk and pancreatitis risk when TG are very high. Lower levels associated with lower residual risk.



Description

Protein that helps clear triglycerides from the blood; imbalance can track with high triglycerides and heart risk. Higher levels associated with high triglycerides. Lower levels associated with rare genetic deficiency and chylomicronemia.



Description

Protein that moves fats and cholesterol; certain forms increase risk for Alzheimer’s and heart disease. Higher levels associated with remnant phenotype and dyslipidemia. Lower levels associated with impaired trafficking. Caveat: does not measure ApoE subtype - requires genetic testing.



Description

Regulates LDL (‘bad’) cholesterol levels; higher activity means higher cholesterol and more artery risk. Lower levels associated with lower risk. Caveat: PCSK9 inhibitors lower serum levels and statins can raise serum levels, epigenetic signal may be altered with these meds.

BLOOD PRESSURE

Vanillic Acid (VAA)_m

Vanillic Acid_m



Description

Vanillic Acid (VAA) is a metabolite derived from phenylalanine metabolism. Elevated levels of VAA are associated with increased cardiovascular disease (CVD) risk and hypertension (HTN) and may indicate pathways contributing to vascular inflammation, oxidative stress, or hormonal regulation of blood pressure. Conversely, low levels of VAA might reflect insufficient phenylalanine metabolism or altered gut microbiome activity, which could indicate disruptions in metabolic or microbial processes.

Phenylacetylglutamine_m



Description

Phenylacetylglutamine (PAG) is a gut-derived metabolite linked to phenylalanine metabolism. Elevated PAG levels are associated with increased cardiovascular and kidney disease risk, systemic inflammation, and reduced longevity.

Recommendations

High-fiber diet to support gut health, limiting red meat and phenylalanine-rich foods (processed meats, dairy), hydration. Exercise and stress management, mindfulness or yoga to reduce systemic inflammation.

- Probiotics
- Prebiotics
- NAC

Dimethylarginine, SDMA and ADMA_m



Description

Markers that block blood vessel relaxation; higher levels are linked to kidney disease and heart risk. Higher levels associated with cardiovascular events and reduced GFR. Lower levels associated with better Nitric Oxide bioavailability and renal handling. Caveat: ACEi, ARB, SGLT2 inhibitors, PDE-5 inhibitors, and statins can change serum faster than epigenetic signal.

Recommendations

Aerobic plus resistance training, Mediterranean diet, manage BP, lipids, and glucose, weight loss if needed

- EPA or DHA
- Cocoa flavanols
- L-citrulline with clinician

METABOLIC MARKERS

HgbA1c_m



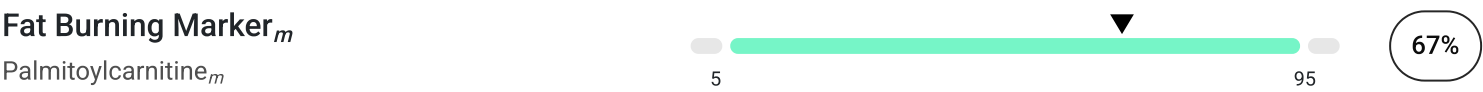
Description

Measure of blood sugar levels over a period of time, indicative of glucose control.



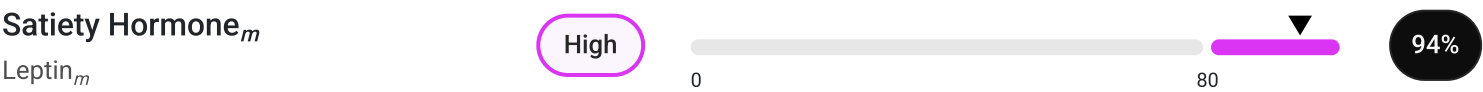
Description

Blood sugar level, related to diabetes risk.



Description

Palmitoylcarnitine plays a role in fatty acid transport into mitochondria for oxidation. High levels can signal impaired fatty acid oxidation, while low levels might affect energy production from fats.



Description

Leptin is a hormone produced by fat cells that helps regulate appetite and energy balance by signaling the brain to reduce hunger and increase energy expenditure. High levels may indicate leptin resistance and excess body fat, while low levels can suggest insufficient fat reserves or impaired metabolic function.

Recommendations

Reduce body fat through a balanced diet rich in whole foods, moderate caloric restriction, and strength training and aerobic activities. Improving sleep quality.

- Berberine
- Omega-3s
- Resveratrol



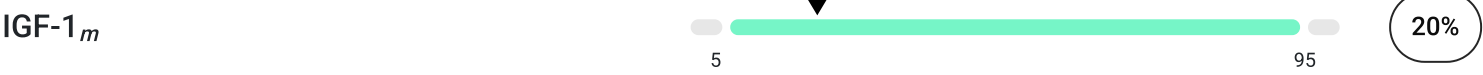
Description

Phenylacetylglutamine (PAG) is a gut-derived metabolite linked to phenylalanine metabolism. Elevated PAG levels are associated with increased cardiovascular and kidney disease risk, systemic inflammation, and reduced longevity.

Recommendations

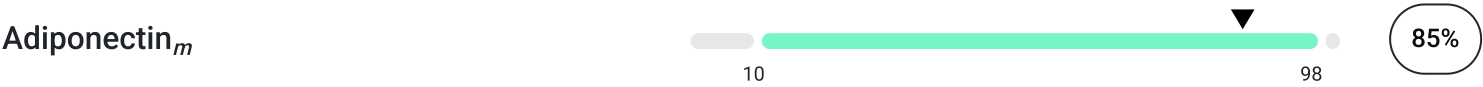
High-fiber diet to support gut health, limiting red meat and phenylalanine-rich foods (processed meats, dairy), hydration. Exercise and stress management, mindfulness or yoga to reduce systemic inflammation.

- Probiotics
- Prebiotics
- NAC



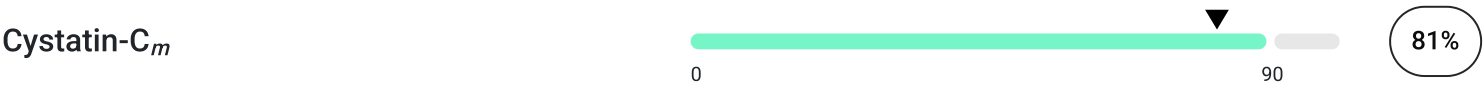
Description

Growth factor that supports muscle and metabolism. Lower levels associated with frailty and sarcopenia. Higher levels associated with cancer risk and acromegaly or GH therapy. Caveat: GH therapy or anabolic steroids can elevate serum while the epigenetic marker may remain lower.



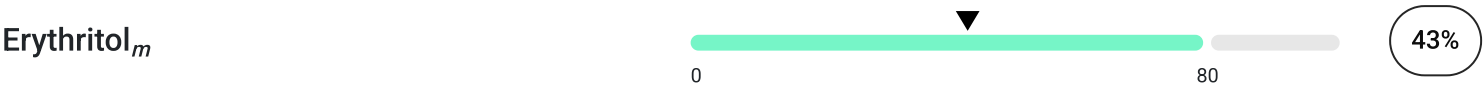
Description

Hormone from fat tissue that improves insulin sensitivity. Lower levels associated with insulin resistance and visceral adiposity. Very high levels associated with heart failure or catabolic states in older adults.



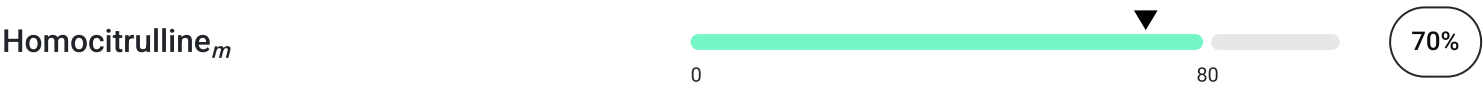
Description

Kidney function measure, not muscle independent like creatinine. Higher levels associated with chronic kidney disease, cardiovascular events, and all-cause mortality. Lower levels associated with preserved filtration.



Description

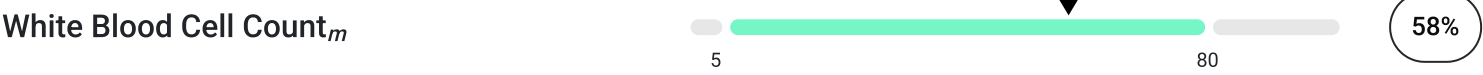
Sugar alcohol made in the body and found in sugar-free foods. Higher levels associated with frequent sugar-free intake and cardiometabolic risk signals in observational studies. Lower levels reflect minimal exposure.



Description

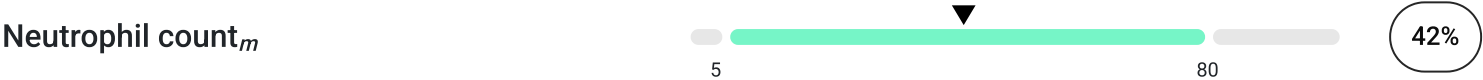
Marker of protein damage from nitrogen waste, often elevated in kidney disease and tied to heart and vessel risk. Higher levels associated with CKD or uremia, atherosclerosis, heart failure, and mortality in CKD cohorts. Lower levels reflect lower carbamylation burden.

IMMUNE MARKERS



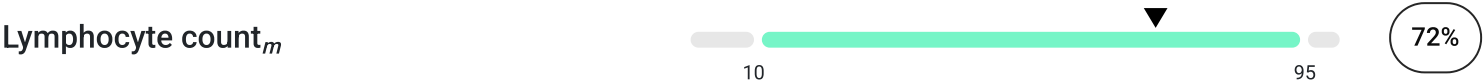
Description

Measures the amount of white blood cells in the blood, important for immune function.



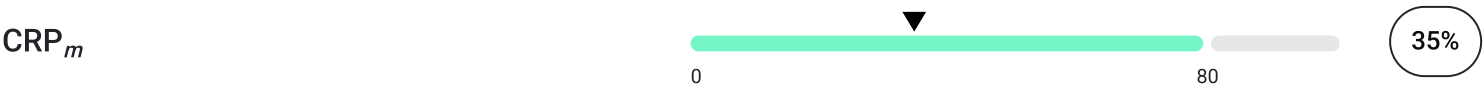
Description

Neutrophil count indicates levels of neutrophils, important for immune defense.



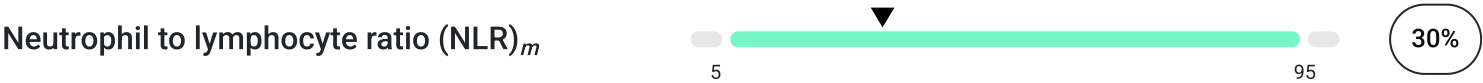
Description

A type of white blood cell, important for immune system function.



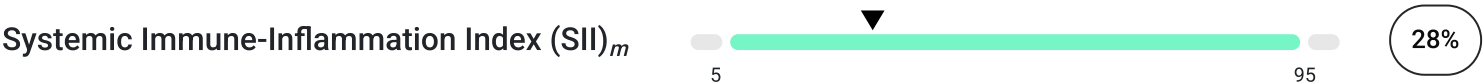
Description

Inflammation marker, linked to cardiovascular and chronic diseases



Description

Neutrophil-to-lymphocyte ratio (NLR) is a marker of systemic inflammation and immune balance. High ratios are associated with chronic inflammation, cardiovascular disease, and poor immune regulation, while low ratios may indicate impaired immune response or hematological abnormalities.



Description

Systemic Immune-Inflammation Index (SII) is a composite marker calculated as (Platelet count × Neutrophil count) / Lymphocyte count. High levels are linked to systemic inflammation, poor prognosis in cancer, and cardiovascular risk, while low levels may indicate immune suppression or reduced inflammatory response.

The CD4/CD8 ratio reflects the balance between helper and cytotoxic T cells, indicating immune system status. High ratios are linked to autoimmune disorders or chronic inflammation, while low ratios suggest immune suppression, aging, or increased risk of infections and certain cancers.

Neurogranin_m

Neurogranin is a calmodulin-binding protein, linked to synaptic plasticity and cognition. High levels can be linked to neurodegenerative diseases (e.g., Alzheimer's), while low levels might impair synaptic plasticity and cognitive function.

Stress management, ensure adequate sleep, engage in moderate exercise (avoiding overtraining).

- 🔗 L-theanine 🔗 Magnesium 🔗 Ashwagandha 🔗 Phosphatidylserine 🔗 Bacopa monnieri

Quinolate_m

Quinolinic acid is a metabolite in the kynurenine pathway linked to neuroinflammation. High levels are linked to neurotoxicity and neurodegenerative disorders, while low levels may impair tryptophan metabolism and affect immune function.

Dopamine 3-O-sulfate_m

Dopamine 3-O-sulfate is a dopamine metabolite involved in neurotransmitter regulation and detoxification. Elevated levels may indicate altered dopamine metabolism, while low levels could suggest reduced detoxification or neurotransmitter imbalance.

Manage stress, get quality sleep, engage in moderate exercise, and reduce stimulants like caffeine.

- L-theanine Magnesium Ashwagandha N-acetylcysteine (NAC) Omega-3s

Cell Repair Marker_m

Transforming growth factor beta (TGF-beta)_m



Description

Transforming Growth Factor Beta (TGF-beta) is a cytokine involved in cell growth, proliferation, differentiation, and apoptosis, with key roles in immune regulation and inflammation. Elevated levels may indicate heightened immune activity or chronic inflammation, while low levels could suggest impaired cellular signaling or immune response.

Brain Anti-inflammatory Protein_m

Progranulin_m



Description

Progranulin is a protein involved in wound healing, inflammation, and neurodegeneration. Elevated levels may reflect active inflammation or tissue repair, while low levels could indicate impaired healing processes or an increased risk of neurodegenerative conditions.

VGF_m

Neurosecretory protein VGF_m



Description

Brain-derived protein that supports mood, memory, and energy balance. Lower levels associated with cognitive decline and depression. Higher levels associated with exercise adaptation. Caveat: may be altered by antidepressants.

INFLAMMATION MARKERS

IL-6_m



Description

IL-6 (Interleukin-6) is a cytokine involved in inflammation and the immune response.

Oxidative Stress Marker_m

Methionine sulfone_m



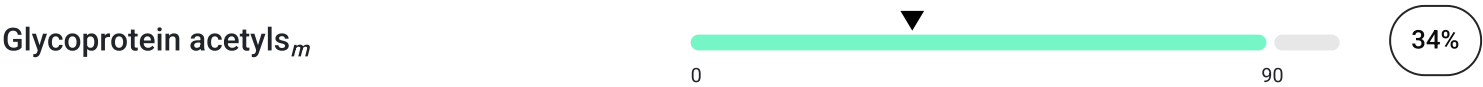
Description

An oxidized form of methionine, a marker of oxidative stress. High levels may indicate excessive oxidative stress or impaired sulfur amino acid metabolism, while low levels could affect protein synthesis and antioxidant production.



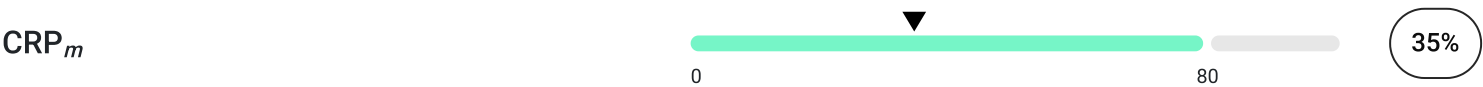
Description

Serum Amyloid A-1 Protein is an acute-phase protein involved in inflammation and immune response. Elevated levels may indicate active inflammation, while low levels could reflect a lack of inflammatory activity.



Description

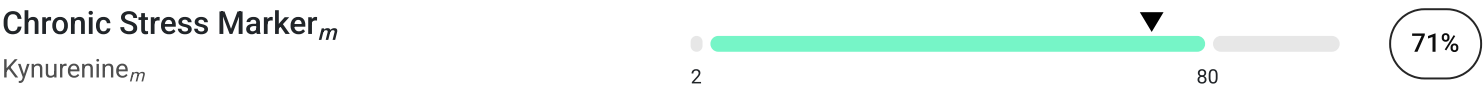
Glycoprotein acetyls (GlycA) are inflammatory biomarkers representing glycosylated acute-phase proteins in the bloodstream. Elevated GlycA levels are associated with systemic inflammation, increased cardiovascular disease risk, and poorer metabolic health, making it a key marker for assessing longevity and chronic disease burden.



Description

Inflammation marker, linked to cardiovascular and chronic diseases

STRESS MARKERS



Description

Kynurenine is a metabolite of tryptophan, involved in immune modulation and neurobiology. High levels are associated with inflammation and neurodegenerative diseases, while low levels may indicate impaired tryptophan metabolism.



Description

VMA is a metabolite of catecholamines (epinephrine and norepinephrine), and high levels are linked to catecholamine excess, while low levels might indicate adrenal insufficiency or low catecholamine turnover.

Recommendations

Reduce sympathetic nervous system activation. Manage stress, ensure adequate sleep, exercise and limit caffeine and stimulant intake.

- L-theanine
- Magnesium
- Ashwagandha
- Phosphatidylserine
- Omega-3s

Long-term adrenal androgen availability. Lower levels associated with frailty, depressed mood, and higher mortality in older adults. Higher levels associated with PCOS or supplementation. Caveat: exogenous DHEA, testosterone, estrogen, or progesterone can make the epigenetic signal appear low despite normal or high serum.

A horizontal progress bar with a light gray track and a purple segment representing 99% completion. A black downward-pointing triangle is positioned above the end of the purple segment. To the right of the bar is a black circle containing the text '99%'. The numbers '0' and '95' are visible at the start and end of the track, respectively.

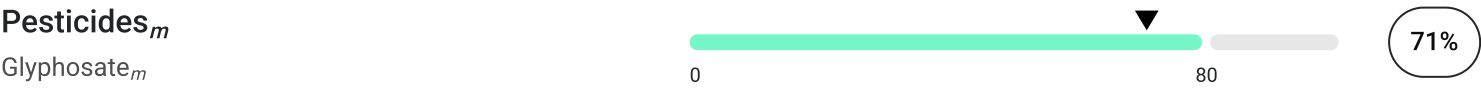
 Glutathione

A horizontal progress bar with a green segment and a grey segment. A black triangle points to the end of the green segment. The number 52% is displayed in a circle to the right of the bar. The bar is labeled with 0 at the start and 95 at the end.

0 95

94%

2-Hydroxyfluorene sulfate is a metabolite that indicates exposure to polycyclic aromatic hydrocarbons (PAHs), which are found in tobacco smoke, grilled or charred foods, and industrial emissions.



Energy Balance Marker_m

Dynamin-1-like protein (DRP1)_m



Description

DRP1 is a protein involved in mitochondrial fission, crucial for cellular energy production and regulation of apoptosis. Elevated levels are linked to mitochondrial fission and could contribute to mitochondrial dysfunction

ATP synthase (Enzyme)_m

ATP synthase subunit beta (ATP5B)_m



Description

A component of the mitochondrial ATP synthase complex, high levels are associated with cellular stress and mitochondrial dysfunction.

OXIDATIVE DEFENSE

Myeloperoxidase (MPO)_m



Description

An enzyme released during inflammation and used as a marker for oxidative stress.

Oxidative Damage Marker_m

Allantoin_m



Description

Allantoin is formed through oxidation of uric acid and thus serves as a marker of oxidative stress. Higher levels of allantoin in the blood increased oxidative stress, inflammation, or other stress-related metabolic disturbances.

Oxidative Stress Indicator_m

Methionine sulfone_m



Description

An oxidized form of methionine, a marker of oxidative stress. High levels may indicate excessive oxidative stress or impaired sulfur amino acid metabolism, while low levels could affect protein synthesis and antioxidant production.

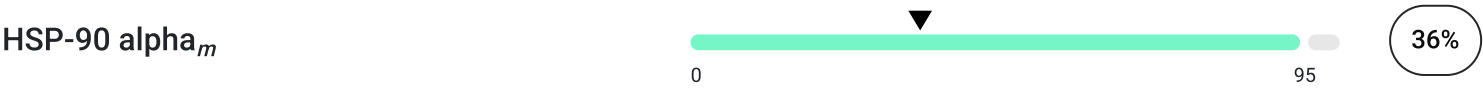
Description

Strain on protein repair and folding systems. Higher levels associated with inflammation, insulin resistance, CKD, and higher cardiometabolic risk. Lower levels reflect lower cumulative damage.

Recommendations

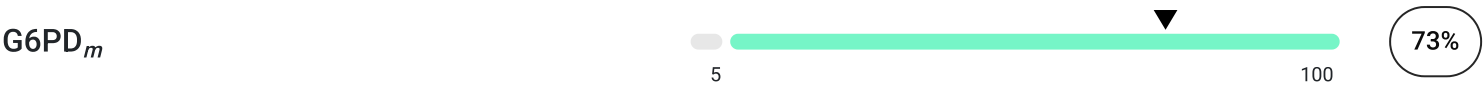
Mediterranean eating, fewer ultra-processed foods and sugars, regular exercise, sleep, tobacco avoidance

- Omega-3s
- Benfotiamine or carnosine with clinician



Description

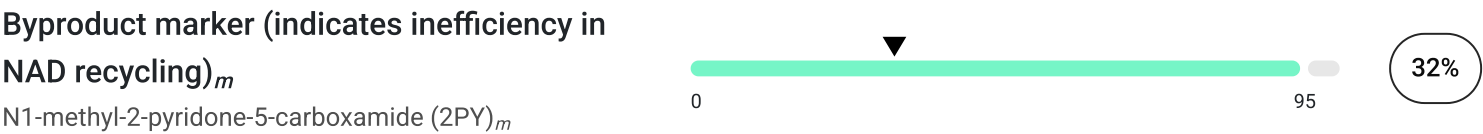
"Chaperone" protein that helps other proteins fold. Higher levels associated with cancer progression, inflammatory states, and adverse prognosis. Lower levels associated with lower chaperone demand.



Description

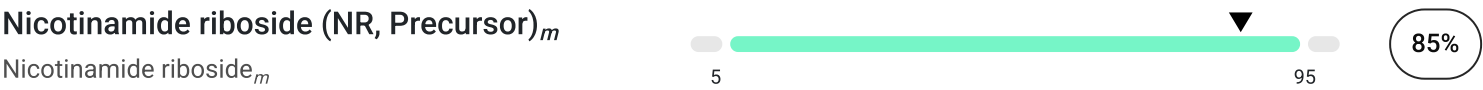
Enzyme that protects red blood cells from stress. Lower levels associated with inherited G6PD deficiency and hemolysis risk. Higher levels generally benign.

NAD+ METABOLISM



Description

N1-methyl-2-pyridone-5-carboxamide (2PY) is a metabolite of NAD+ that can accumulate with increased NAD+ production. Elevated 2PY levels may inhibit enzymes like PARP, potentially impairing DNA repair and cellular energy metabolism, which could negatively affect longevity.



Description

Nicotinamide Riboside is a precursor to NAD+. Low levels can lead to reduced NAD+ production and decreased cellular energy, while high levels of NR indicate efficient NAD+ synthesis but could suggest over-reliance on supplementation.

Nicotinamide (Precursor)_m

Nicotinamide_m



Description

Nicotinamide is a form of vitamin B3, essential for NAD+ production and cellular energy. Low levels are associated with lower NAD+ production, and high levels are associated with inhibition of Sirtuins and NAD+ recycling pathways.

KETONES

Beta Hydroxybutyrate_m

3-hydroxybutyrate (BHBA)_m



Description

3-Hydroxybutyrate (BHBA) is a ketone body produced during fat metabolism, primarily in the liver, and serves as an alternative energy source for the brain and muscles, especially during periods of low carbohydrate intake or fasting.

SUPPLEMENTS

Alpha-ketoglutarate_m



Description

Alpha-ketoglutarate is a key intermediate in the Krebs cycle, crucial for energy production, amino acid metabolism, and cellular function regulation.

Spermidine_m



Description

Spermidine is a polyamine compound involved in cellular growth, proliferation, and apoptosis, known for its potential role in promoting autophagy and longevity, and is found in foods like aged cheese, soy products, and whole grains.